TEMPEST: Porewater SO4/Cl

202606 24-81 Samples

2026-09-02

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##Add Required Packages

0.1 Run Information

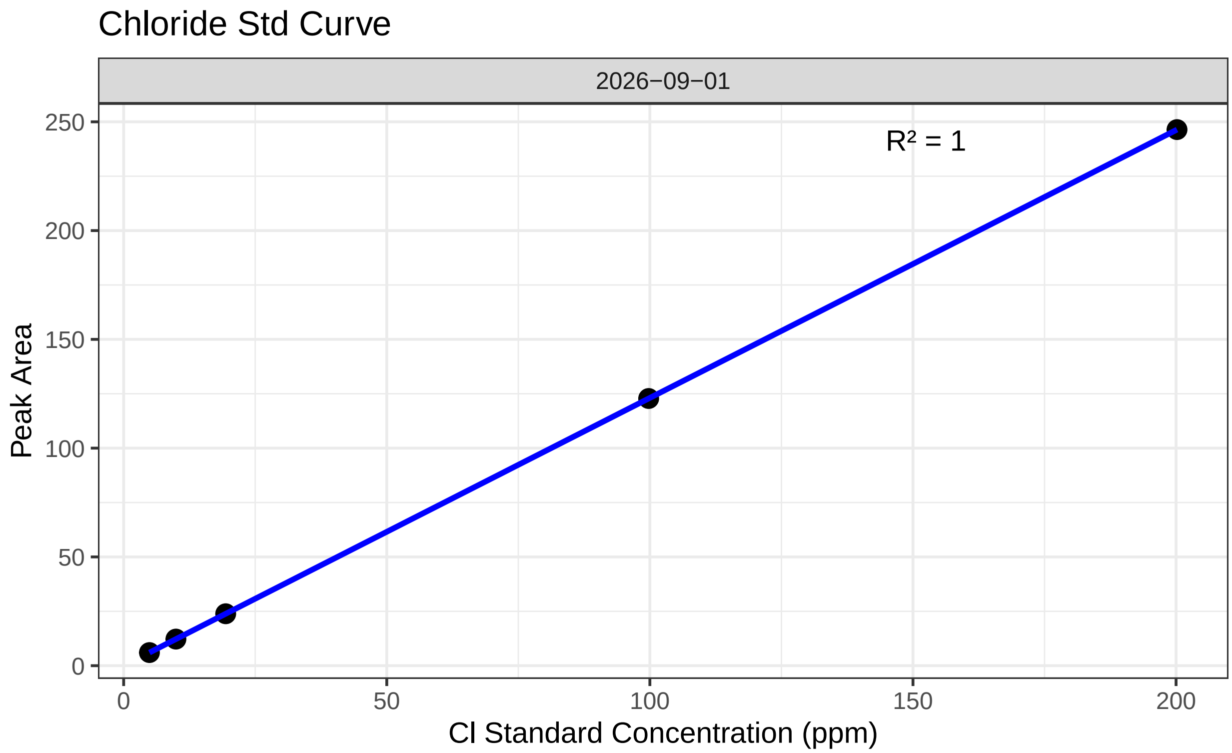
```
###### Run information - PLEASE CHANGE
  Sample_Year = "2026"
  Date_Run = "2026-09-01"  #Date that instrument was run
  Run_by = "Zoe Read"  #Instrument user
  Script_run_by = "Zoe Read" #Code user
  run_notes = "
  " #any notes from the run

##Sample data that was entered incorrectly
# The Old ID is the original, incorrecly-entered ID and the New ID is the correct ID to change it to.
Old_ID_1 = NA
New_ID_1 = NA

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
  raw_file_name_cl = "Raw Data/COMPASS_TEMPEST_202606_24-81_Cl.txt"
  raw_file_name_so4 = "Raw Data/COMPASS_TEMPEST_202606_24-81_SO4.txt"

#Sample ID Path
  SampleID_path = "Raw Data/TEMPEST_SO4_20260501-20260612_corrected.csv"

#file path and name of processed data file
  processed_file_name = "Processed Data/COMPASS_TEMPEST_Processed_Cl_SO4_202606_24-81.csv"

###### Log Files - PLEASE CHECK

#qaqc log file path for this year copied over from COMPASS GitHub
  Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2026.csv"
```

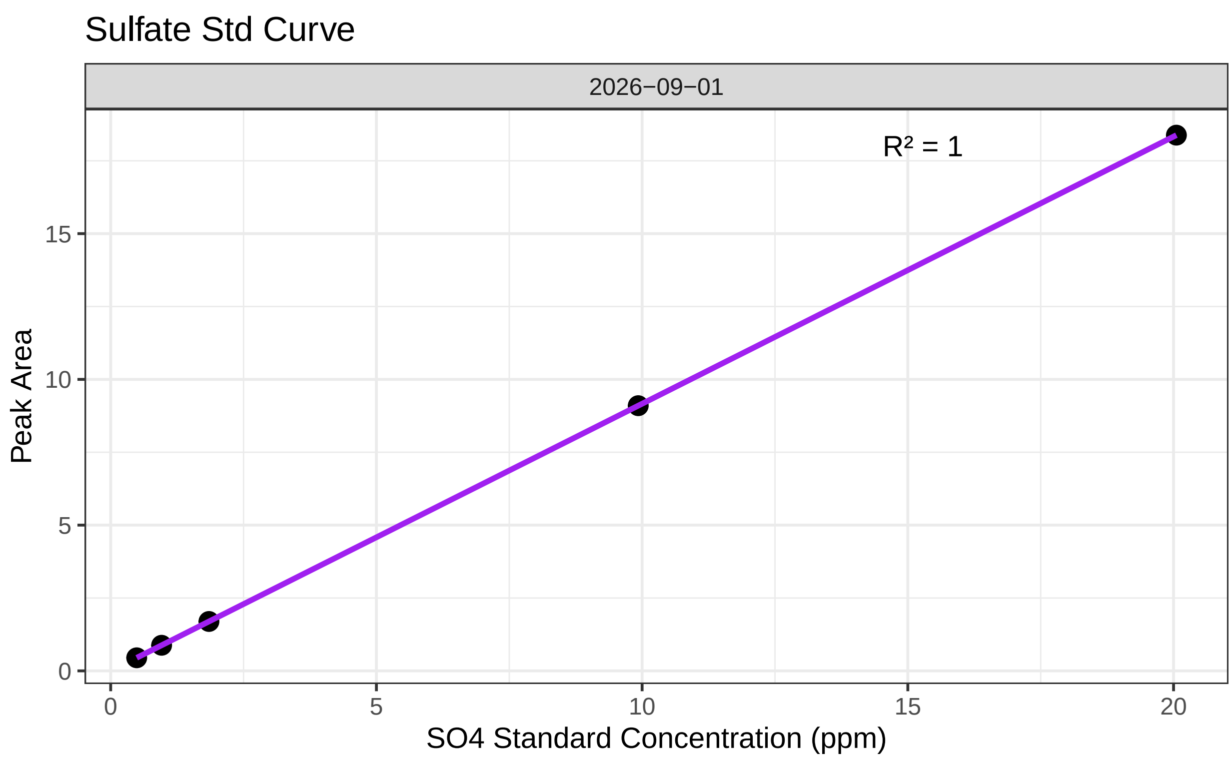
##Set Up Code - constants and QAQC cutoffs

##Import Sample Data

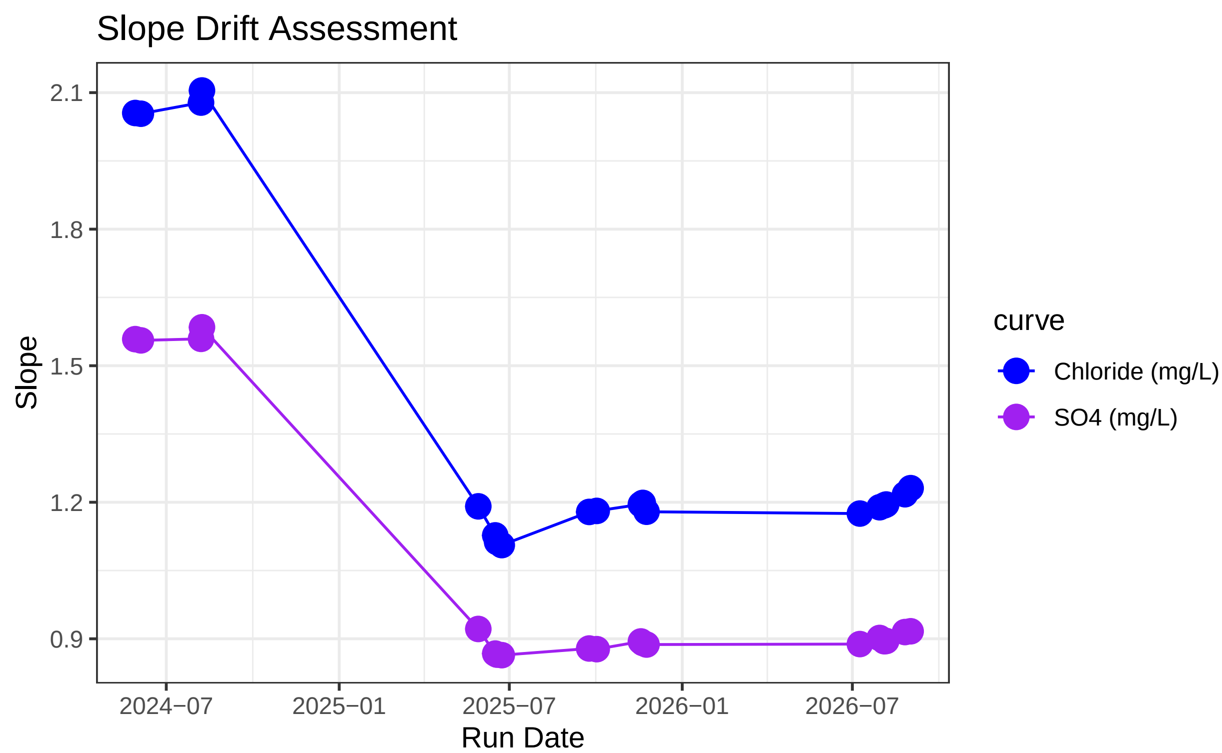
##Fix Sample IDs entered wrong

0.2 Assess Standard Curves






```
## [1] "QAQC log file exists and has been read into the code."
```




```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

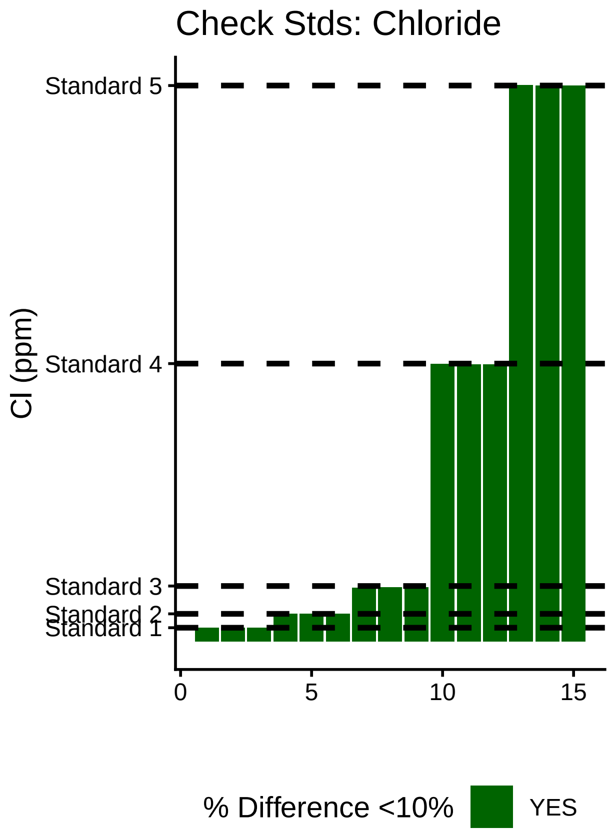
0.3 Assess Check Standards

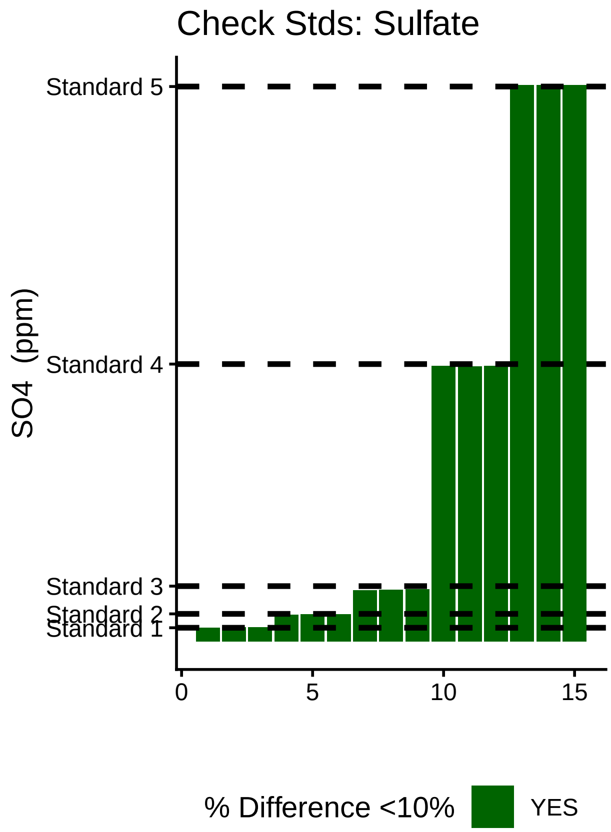
```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl   sd_Cl     cv_Cl flag_Cl
##   <chr>        <dbl>   <dbl>     <dbl> <chr>
## 1 Standard 1    4.90 0.00987 0.00201  Chloride Check Standard RSD within Range ~
## 2 Standard 2    9.93 0.00555 0.000560 Chloride Check Standard RSD within Range ~
## 3 Standard 3   19.5  0.0655  0.00336  Chloride Check Standard RSD within Range ~
## 4 Standard 4   99.7  0.0892  0.000894 Chloride Check Standard RSD within Range ~
## 5 Standard 5  200.   0.146   0.000732 Chloride Check Standard RSD within Range ~
```

```
## # A tibble: 5 x 5
##   sample_ID  mean_SO4  sd_SO4    cv_SO4 flag_SO4
##   <chr>         <dbl>   <dbl>     <dbl> <chr>
## 1 Standard 1    0.508 0.0157  0.0310    Sulfate Check Standard RSD within Range~
## 2 Standard 2    0.979 0.0184  0.0188    Sulfate Check Standard RSD within Range~
## 3 Standard 3    1.87  0.0200  0.0107    Sulfate Check Standard RSD within Range~
## 4 Standard 4    9.92  0.0102  0.00103   Sulfate Check Standard RSD within Range~
## 5 Standard 5   20.1   0.00121 0.0000605 Sulfate Check Standard RSD within Range~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```





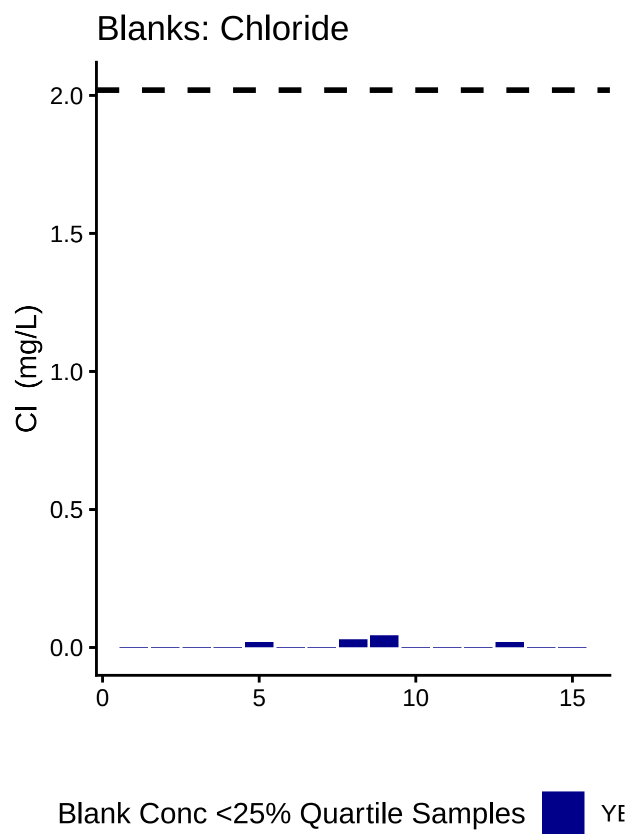

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

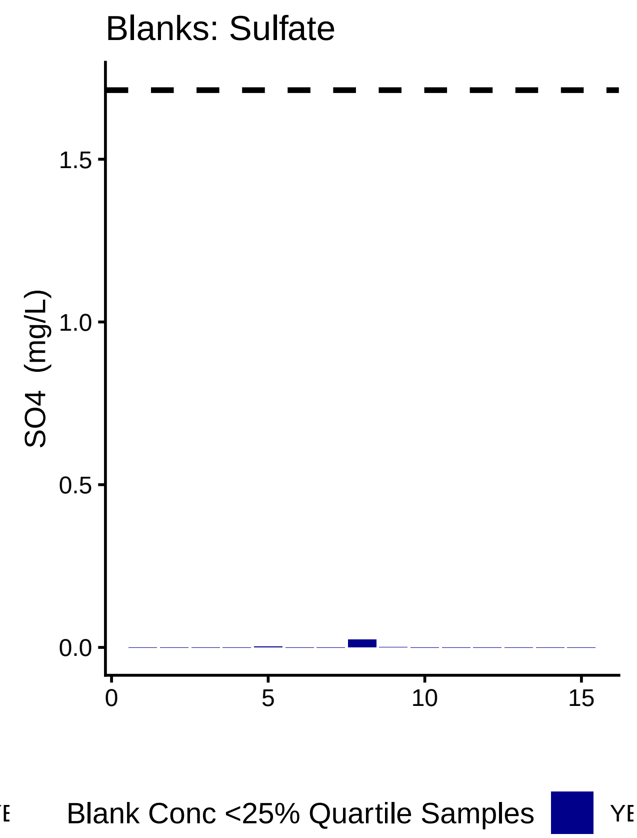
```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

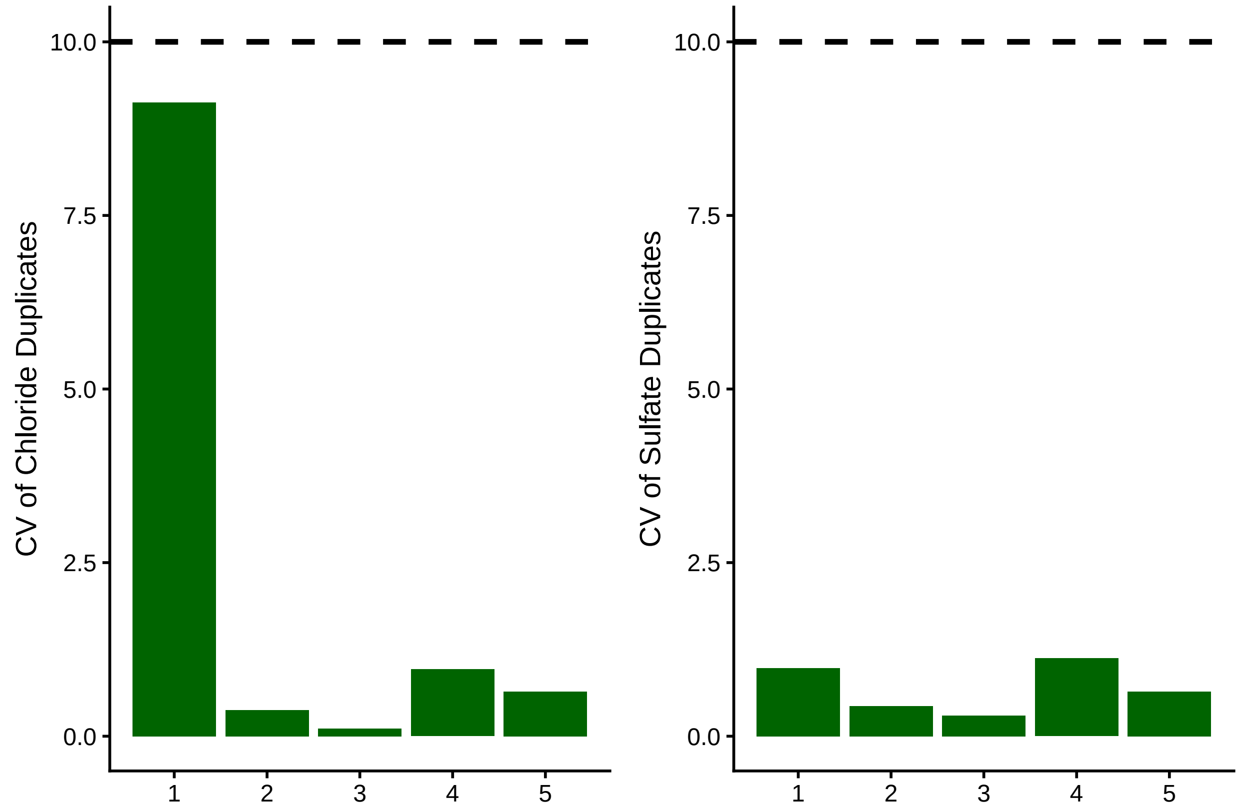
```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```





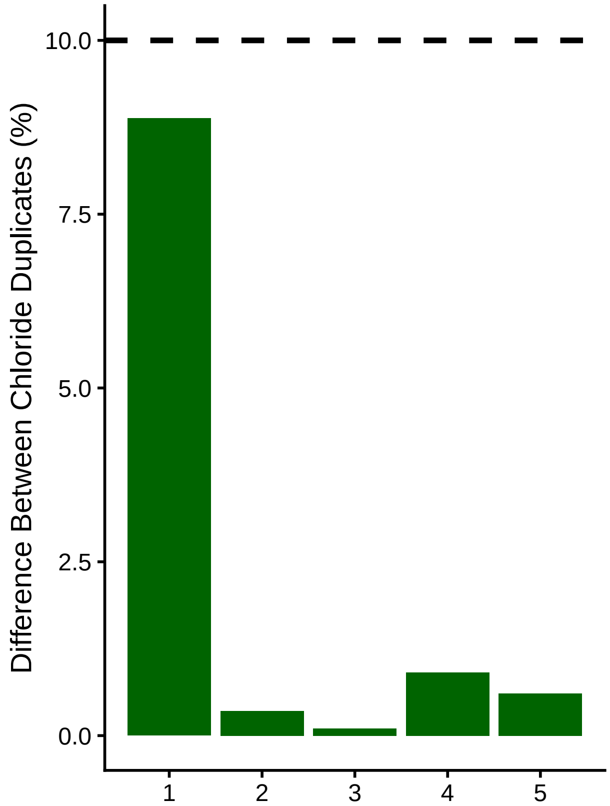

```
## Chloride blanks mean ppm:

## [1] 0.00728

## Sulfate blanks mean ppm:

## [1] 0.001833333
```

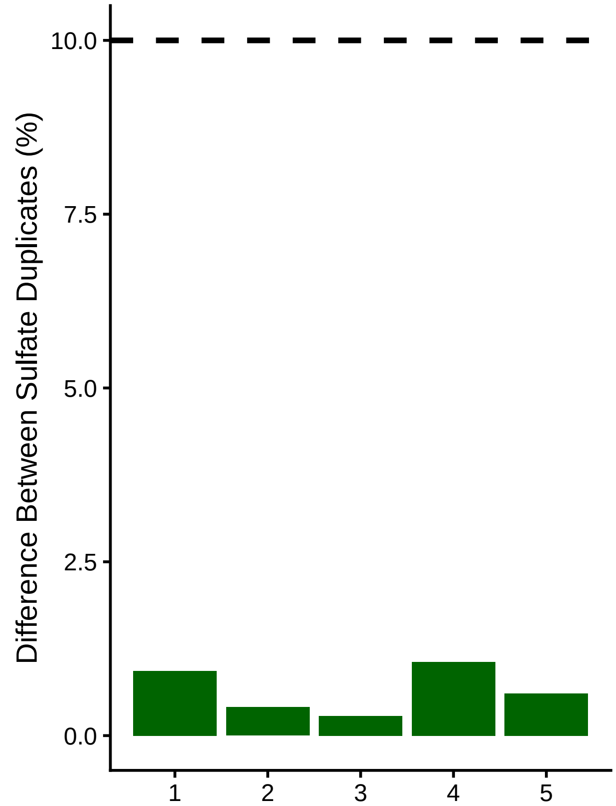
0.5 Assess Duplicates




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```





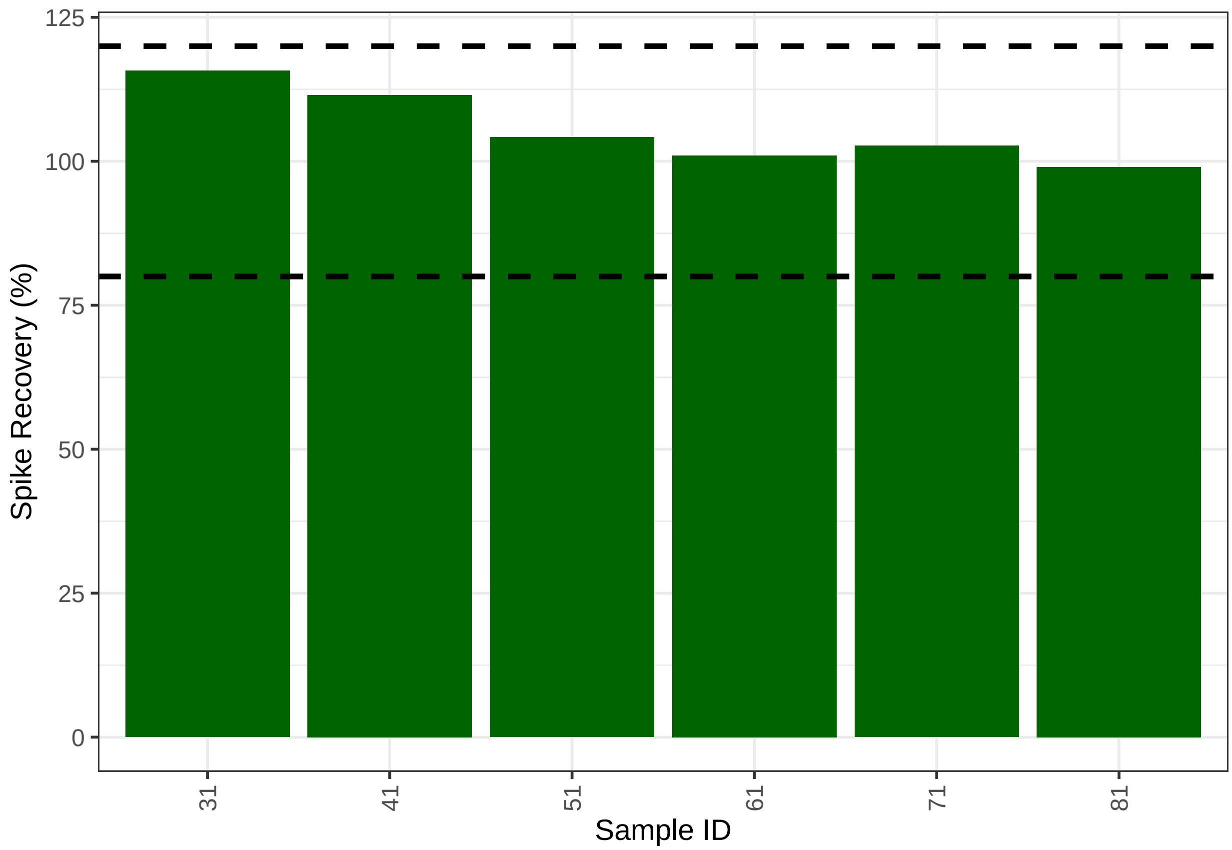

```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"

## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for TEMPEST this depends on the sample so...
all_dat$Dilution <- 50
all_dat$Dilution <-  ifelse(str_detect(all_dat$sample_ID, "TMP"), 50, all_dat$Dilution)
all_dat$Dilution <-  ifelse(str_detect(all_dat$sample_ID, "EST_SourceWater"), 100, all_dat$Dilution)
all_dat$Dilution <-  ifelse(str_detect(all_dat$sample_ID, "SW_SourceWater"), 100, all_dat$Dilution)
```

```r
all_dat$Dilution <-  ifelse(str_detect(all_dat$sample_ID, "FATE"), 100, all_dat$Dilution)

# head(all_dat)
```

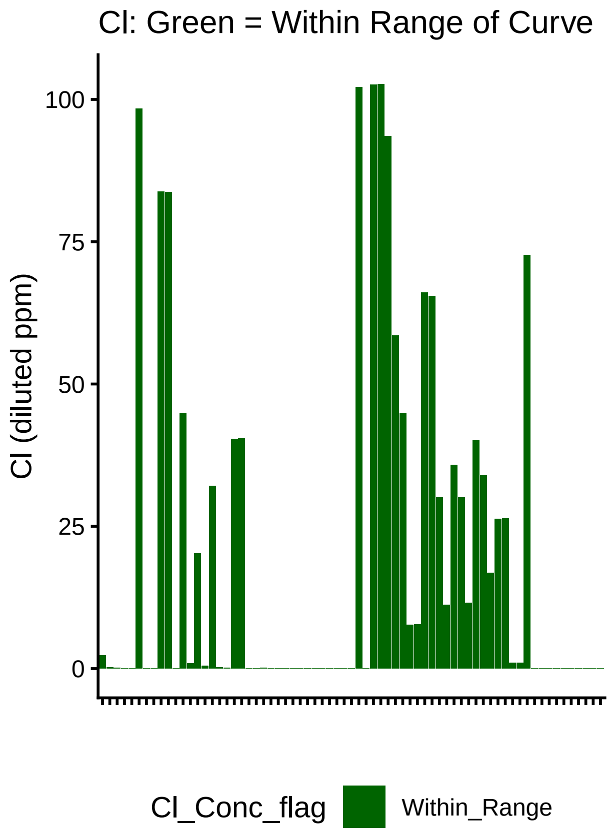
0.7 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```



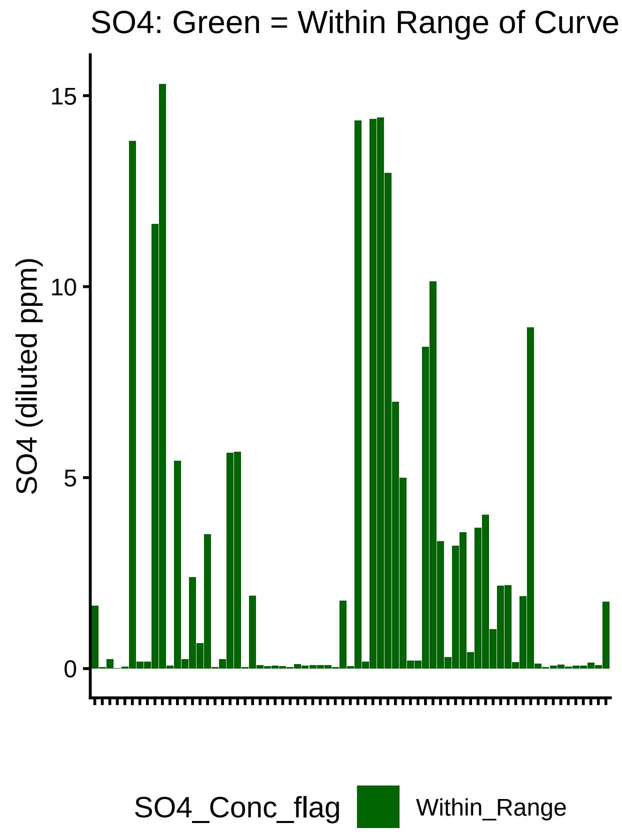


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

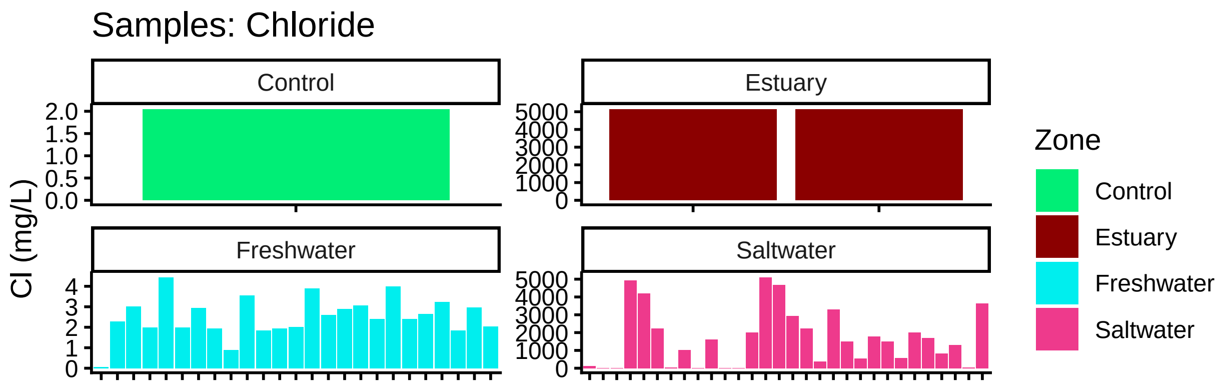
##Merge with Sample ID list to get actual sample IDs

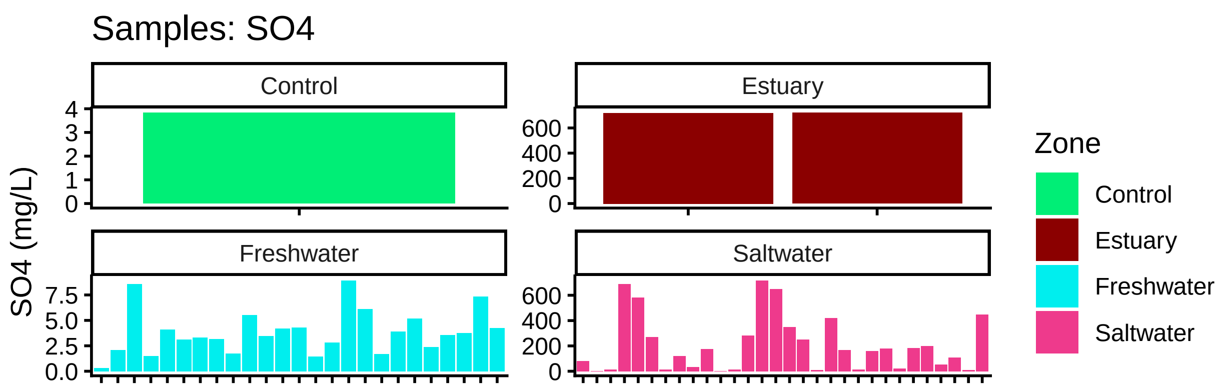
##Pull in active porewater inventory

##Check samples against metadata

```
## ***All sample IDs are present in metadata.***
```

0.9 Visualize Data by Plot





0.10 Export Processed Data

#end